

S&S End Sem Exam Solution 2025

Sol(1):

Given signals,

$$x(t) = \begin{cases} t+1 & 0 \leq t \leq 1 \\ 2-t & 1 < t \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

$$h(t) = \delta(t+2) + 2\delta(t+1)$$

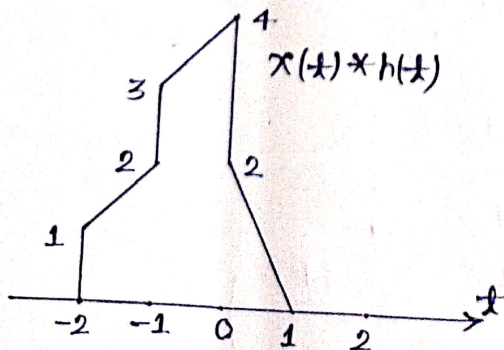
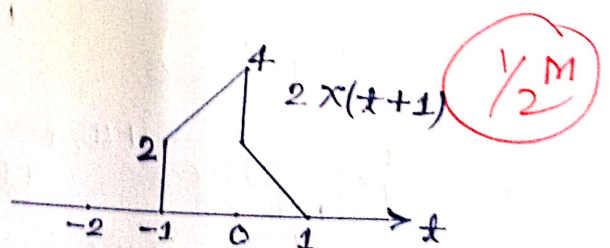
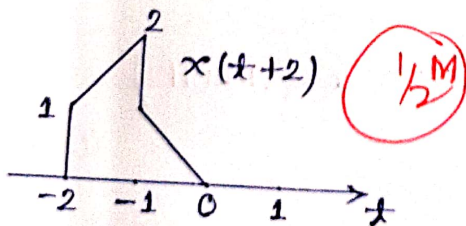
$$x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau) d\tau = \int_{-\infty}^{\infty} x(t-\tau) h(\tau) d\tau$$

$$= \int_{-\infty}^{\infty} x(\tau) \{ \delta(t-\tau+2) + 2\delta(t-\tau+1) \} d\tau$$

$$= \int_{-\infty}^{\infty} x(\tau) \cdot \delta(t-\tau+2) d\tau + 2 \int_{-\infty}^{\infty} x(\tau) \delta(t-\tau+1) d\tau$$

$$= x(t+2) \int_{-\infty}^{\infty} \delta(t-\tau+2) d\tau + 2 x(t+1) \int_{-\infty}^{\infty} \delta(t-\tau+1) d\tau$$

$$= x(t+2) + 2 x(t+1)$$

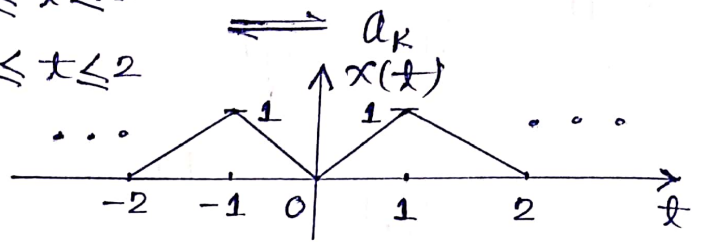


$$x(t) * h(t) = \begin{cases} t+3 & -2 < t \leq -1 \\ t+4 & -1 < t \leq 0 \\ 2-2t & 0 < t \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

(marked $\frac{1}{2}M$)

Sol(2): Given periodic signal ($T=2$),

$$x(t) = \begin{cases} t & 0 \leq t \leq 1 \\ 2-t & 1 \leq t \leq 2 \end{cases}$$



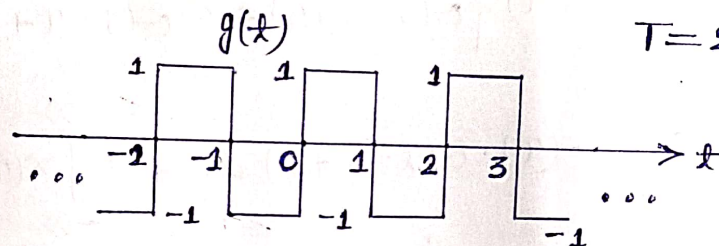
$$(a) \quad a_0 = \frac{1}{T} \int_T x(t) dt$$

$$= \frac{1}{2} \int_0^1 t dt + \frac{1}{2} \int_1^2 (2-t) dt = \frac{1}{2} \left(\frac{t^2}{2} \right)_0^1 + \frac{1}{2} \left(2t - \frac{t^2}{2} \right)_1^2$$

$$= \frac{1}{2} \left(\frac{1}{2} - 0 \right) + \frac{1}{2} \left(4 - 2 - 2 + \frac{1}{2} \right) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

→ (1 MARK)

$$(b) \quad \text{Let } g(t) = \frac{dx(t)}{dt} \Rightarrow b_k, \quad g(t) = \sum_{k=-\infty}^{\infty} b_k e^{jk\omega_0 t}$$



$$T=2, \quad \omega_0 = \frac{2\pi}{T}$$

$$= \pi \text{ rad/sec}$$

↓
(0.5 M)

$$b_0 = \frac{1}{T} \int_T g(t) dt$$

$$= \frac{1}{2} \int_0^1 (+1) dt + \frac{1}{2} \int_1^2 (-1) dt = +\frac{1}{2} \left(t \right)_0^1 - \frac{1}{2} \left(t \right)_1^2$$

$$= \frac{1}{2} - \frac{1}{2} = 0$$

→ (0.5 MARK)

$$b_k = \frac{1}{T} \int_T g(t) \cdot e^{-jk\omega_0 t} dt$$

$$\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi$$

$$= \frac{1}{2} \int_0^1 (+1) e^{-jk\pi t} dt + \frac{1}{2} \int_1^2 (-1) e^{-jk\pi t} dt$$

$$= \frac{1}{2} \times \left(\frac{e^{-jk\pi t}}{-jk\pi} \right)_0^1 - \frac{1}{2} \times \left(\frac{e^{-jk\pi t}}{-jk\pi} \right)_1^2$$

$$= \frac{1}{2} \left(\frac{e^{-jk\pi} - 1}{-jk\pi} \right) - \frac{1}{2} \left(\frac{e^{-j2k\pi} - e^{-jk\pi}}{-jk\pi} \right)$$

$$= \frac{1}{jk\pi} (1 - e^{-jk\pi})$$

→ (1.0 Mark)

(C) Differentiation property of CT fourier series

$$\text{If } x(t) \xrightarrow{\text{FS}} a_k$$

$$\text{then } \frac{d}{dt} x(t) \xrightarrow{\text{FS}} jk\omega_0 a_k$$

→ (1 Mark)

$$\therefore y(t) = \frac{d}{dt} x(t)$$

$$\therefore b_k = jk\omega_0 a_k = jk\pi a_k$$

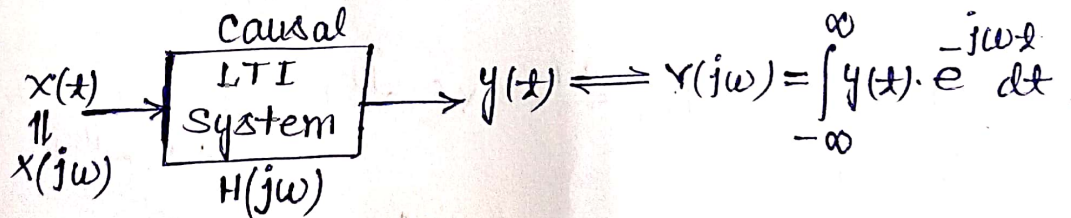
$$\text{Hence, } a_k = \frac{1}{jk\pi} b_k$$

$$= \frac{1}{jk\pi} \times \frac{1}{jk\pi} (1 - e^{-jk\pi})$$

$$= -\frac{1}{k^2\pi^2} (1 - e^{-jk\pi})$$

→ (1 Mark)

Sol(3):



$$\text{Given that - } H(j\omega) = \left(\frac{1}{j\omega + 3} \right) \quad \text{--- (1)}$$

$$\& \quad y(t) = e^{-3t} u(t) - e^{-4t} u(t)$$

$$\therefore Y(j\omega) = \int_{-\infty}^{\infty} e^{-3t} u(t) \cdot e^{-j\omega t} dt - \int_{-\infty}^{\infty} e^{-4t} u(t) e^{-j\omega t} dt$$

$$= \int_0^{\infty} e^{-(j\omega + 3)t} \cdot dt - \int_0^{\infty} e^{-(j\omega + 4)t} \cdot dt$$

$$= \left[\frac{e^{-(j\omega + 3)t}}{-(j\omega + 3)} \right]_0^{\infty} - \left[\frac{e^{-(j\omega + 4)t}}{-(j\omega + 4)} \right]_0^{\infty}$$

$$\begin{aligned}
 &= \frac{0-1}{-(j\omega+3)} - \frac{0-1}{-(j\omega+4)} \\
 &= \frac{1}{3+j\omega} - \frac{1}{4+j\omega} \\
 &= \frac{1}{(3+j\omega)(4+j\omega)} \quad \text{--- (2)}
 \end{aligned}$$

→ (3 Marks)

We know that -

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)}$$

$$\begin{aligned}
 \therefore X(j\omega) &= \frac{Y(j\omega)}{H(j\omega)} = \frac{1/(3+j\omega)(4+j\omega)}{1/(3+j\omega)} \\
 &= \frac{1}{(4+j\omega)} \quad \text{--- (3)}
 \end{aligned}$$

→ (2 Marks)

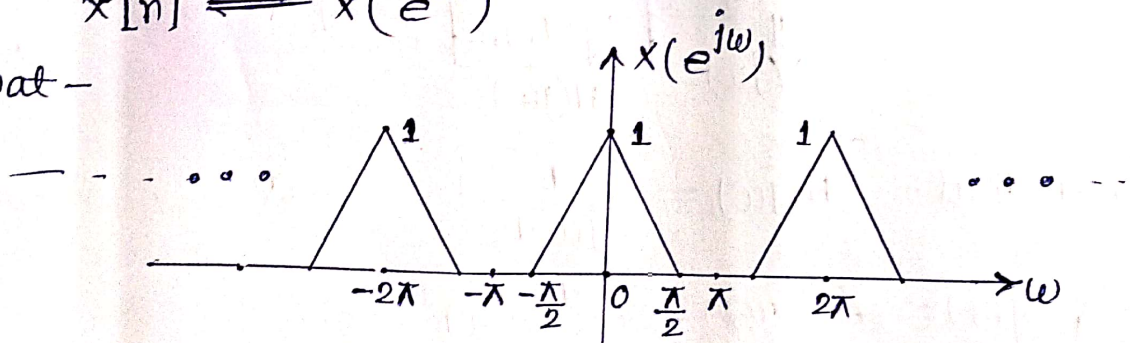
Taking the inverse Fourier transform of $X(j\omega)$, we have -

$$x(t) = e^{-4t} u(t)$$

→ (1 Mark)

Sol (4): $x[n] \xrightarrow{\text{DFT}} x(e^{j\omega})$

Given that -

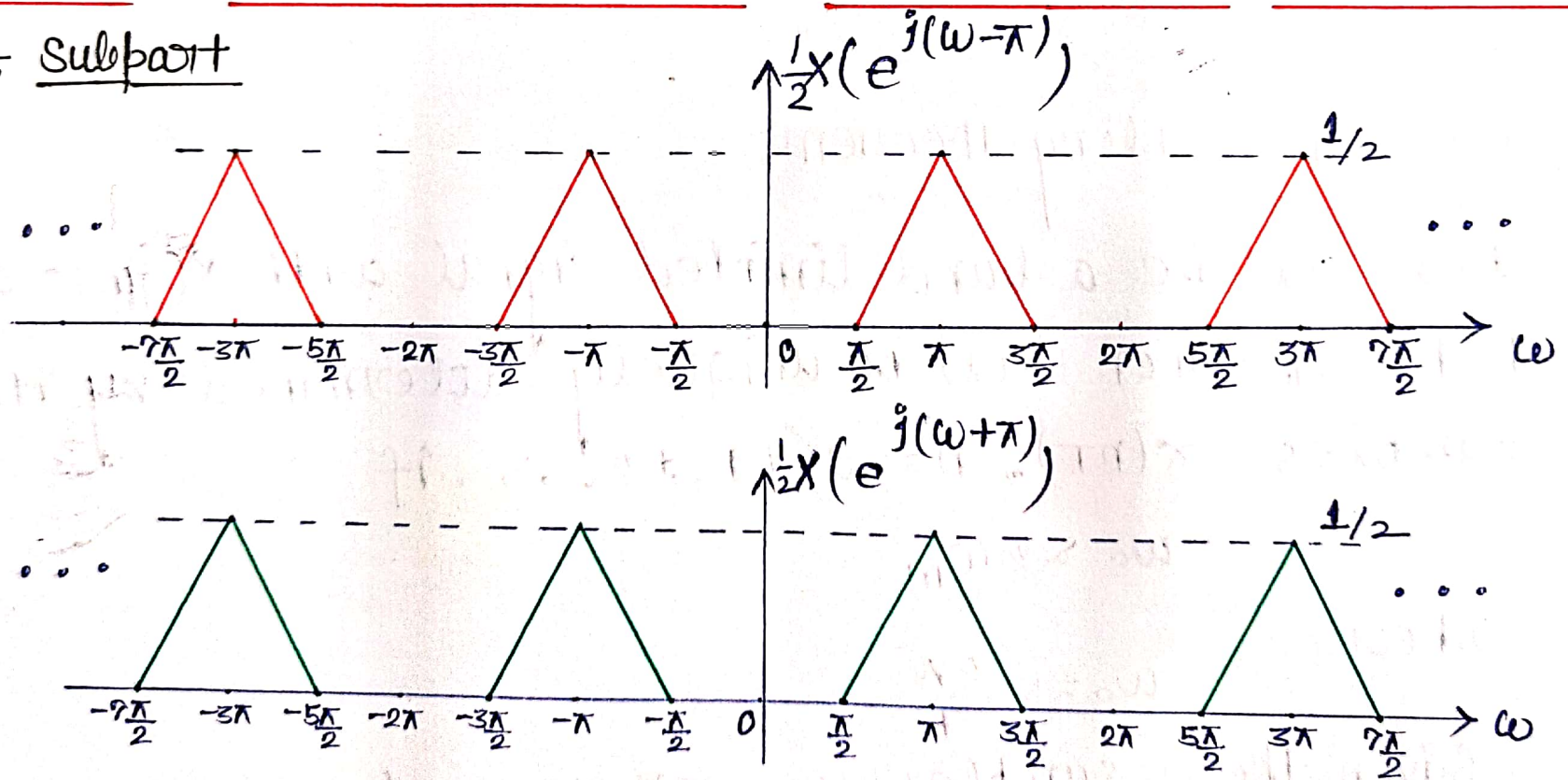


$$v[n] = x[n] \cdot b[n] \quad \text{--- (1)}$$

$$(i) \quad b[n] = \cos \pi n = \left(\frac{e^{j\pi n} + e^{-j\pi n}}{2} \right), \quad \omega_0 = \pi$$

$$\begin{aligned}
 1 &\xrightarrow{\text{DFT}} 2\pi \delta(\omega) \\
 e^{j\omega_0 n} &\xrightarrow{\text{DFT}} 2\pi \delta(\omega - \omega_0) \quad \text{--- (2)}
 \end{aligned}$$

Sol(4): Subpart



Let $p[n] \xrightarrow{\text{DFT}} P(e^{j\omega})$

$$\therefore P(e^{j\omega}) = \frac{1}{2} \{ 2\pi \delta(\omega - \pi) + 2\pi \delta(\omega + \pi) \}$$

$$= \pi \{ \delta(\omega - \pi) + \delta(\omega + \pi) \} \quad \text{--- (3)}$$

→ (1 Mark)

$$\therefore v[n] = x[n] \cdot p[n] \quad (\text{Given})$$

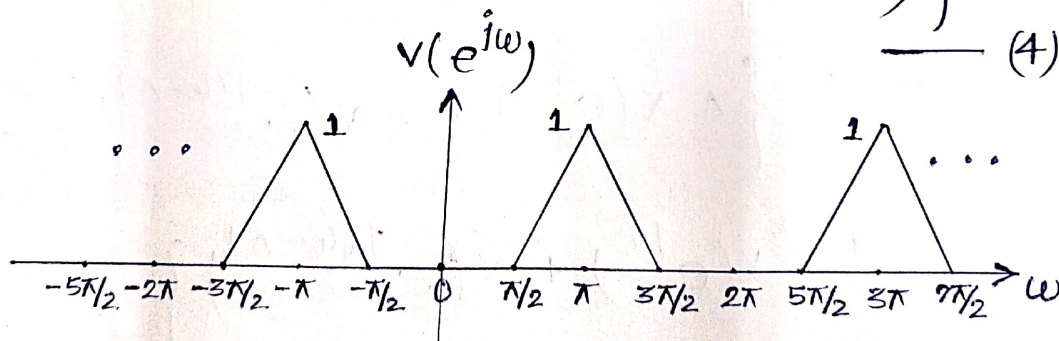
$$\therefore V(e^{j\omega}) = \frac{1}{2\pi} \{ X(e^{j\omega}) * P(e^{j\omega}) \}$$

$$= \frac{1}{2\pi} \left\{ X(e^{j\omega}) * \{ \pi \delta(\omega - \pi) + \pi \delta(\omega + \pi) \} \right\}$$

$$= \frac{1}{2\pi} \left\{ \pi \cdot X(e^{j(\omega - \pi)}) + \pi \cdot X(e^{j(\omega + \pi)}) \right\}$$

$$= \frac{1}{2} \left\{ X(e^{j(\omega - \pi)}) + X(e^{j(\omega + \pi)}) \right\} \quad \text{--- (4)}$$

→ (1 Mark)



→ (1 Mark)

$$(II) p[n] = \sum_{k=-\infty}^{\infty} \delta[n - 2k], \quad N=2$$

$$= \sum_{l=\langle N \rangle} a_l \cdot e^{jl\omega_0 n}$$

$p[n]$ is periodic with period $N=2$ & $\omega_0 = \frac{2\pi}{N} = \pi$

$$\begin{aligned} 1 &\xrightarrow{\text{DFT}} 2\pi \delta(\omega) \\ e^{jl\omega_0 n} &\xrightarrow{\text{DFT}} 2\pi \delta(\omega - l\omega_0) \end{aligned} \quad \text{--- (5)}$$

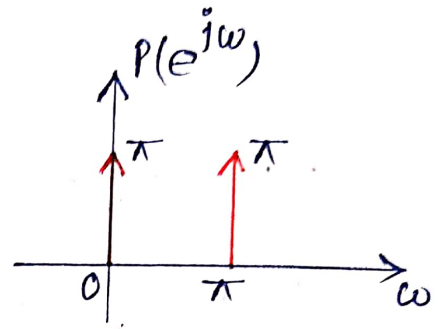
$$\therefore a_l = \frac{1}{N} \sum_{n=\langle N \rangle} \delta[n] \cdot e^{-j l \omega_0 n}$$

$$= \frac{1}{2} \sum_{n=\langle 2 \rangle} \delta[n] e^{-j l \pi n} = \frac{1}{2} \quad \text{--- (6)}$$

$$\therefore P(e^{j\omega}) = \sum_{l=\langle 2 \rangle} \frac{1}{2} \cdot 2\pi \delta(\omega - \pi l)$$

$$= \sum_{l=\langle 2 \rangle} \pi \delta(\omega - \pi l)$$

$$= \pi [\delta(\omega) + \delta(\omega - \pi)] \quad \text{--- (7)}$$



→ (1 MARK)

$$\therefore v[n] = x[n] \cdot p[n] \quad (\text{Given})$$

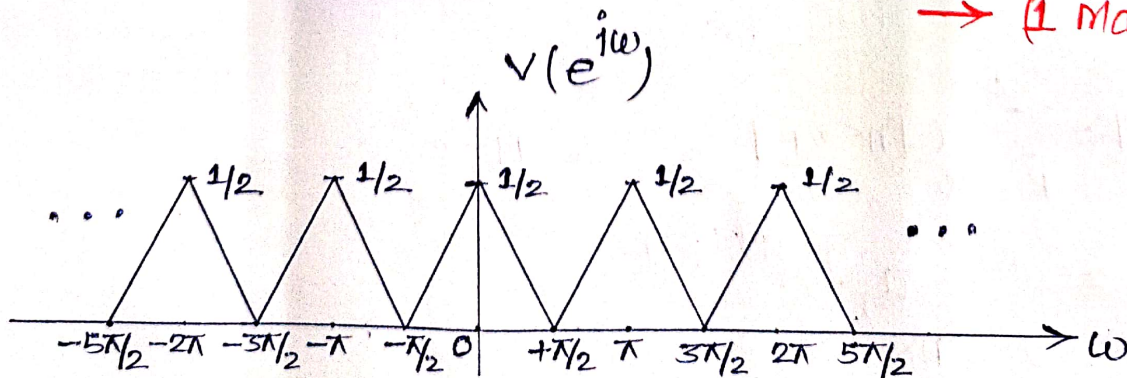
$$\therefore v(e^{j\omega}) = \frac{1}{2\pi} \{ x(e^{j\omega}) * P(e^{j\omega}) \}$$

$$= \frac{1}{2\pi} \{ x(e^{j\omega}) * \{ \pi [\delta(\omega) + \delta(\omega - \pi)] \} \}$$

$$= \frac{1}{2\pi} \{ \pi x(e^{j\omega}) + \pi x(e^{j(\omega - \pi)}) \}$$

$$= \frac{1}{2} \{ x(e^{j\omega}) + x(e^{j(\omega - \pi)}) \} \quad \text{--- (8)}$$

→ (1 MARK)



→ (1 MARK)

Sol (5):

$$x(t) \xrightarrow{LT} X(s)$$

From fact (2), $x(s)$ has 4 poles and no zeros in the finite s -plane. Hence we may assume that $x(s)$ is of the form,

$$x(s) = \frac{A}{(s-a)(s-b)(s-c)(s-d)} \quad \text{--- (1)}$$

→ (1 Mark)

From fact (1), $x(t)$ is real, the poles of $x(s)$ must occur in conjugate reciprocal pairs. Therefore we may assume that $b = a^*$ and $d = c^*$. This results in -

$$x(s) = \frac{A}{(s-a)(s-a^*)(s-c)(s-c^*)} \quad \text{--- (2)}$$

→ (1 Mark)

From fact (1), $x(t)$ is even. Hence the Laplace transform $x(s)$ must also be even. This implies that the poles have to be symmetric about the $j\omega$ -axis. Therefore we may assume that $c = -a^*$. This results in -

$$x(s) = \frac{A}{(s-a)(s-a^*)(s+a^*)(s+a)} \quad \text{--- (3)}$$

→ (1 Mark)

From fact (3), location of pole at $s = \frac{1}{2} e^{j\pi/4}$

$$\begin{aligned} x(s) &= \frac{A}{(s - \frac{1}{2} e^{j\pi/4})(s - \frac{1}{2} e^{-j\pi/4})(s + \frac{1}{2} e^{-j\pi/4})(s + \frac{1}{2} e^{j\pi/4})} \\ &= \frac{A}{(s^2 - \frac{s}{\sqrt{2}} + \frac{1}{4})(s^2 + \frac{s}{\sqrt{2}} + \frac{1}{4})} \quad \text{--- (4)} \end{aligned}$$

→ (1 Mark)

From fact (4),

$$\int_{-\infty}^{\infty} x(t) dt = 4$$

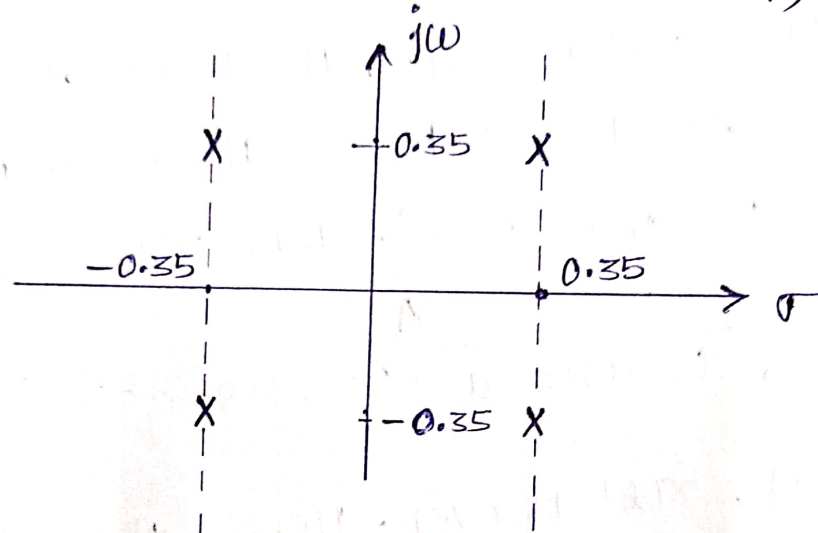
$$X(s=0) = 4 \quad \text{--- (5)}$$

By eqⁿ(4) & eqⁿ(5), we get -

$$A = (1/4)$$

→ (1 MARK)

Therefore,
$$X(s) = \frac{(1/4)}{(s^2 - \frac{s}{\sqrt{2}} + \frac{1}{4})(s^2 + \frac{s}{\sqrt{2}} + \frac{1}{4})}$$



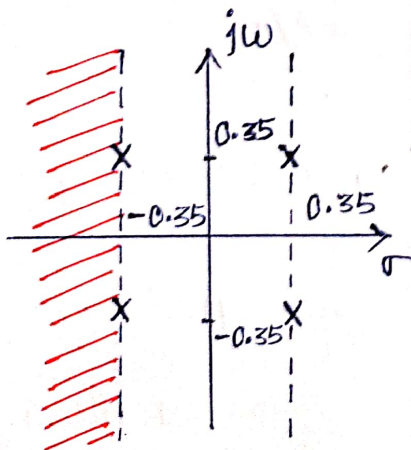
Possible ROC,

$$\operatorname{Re}\{s\} < -0.35$$

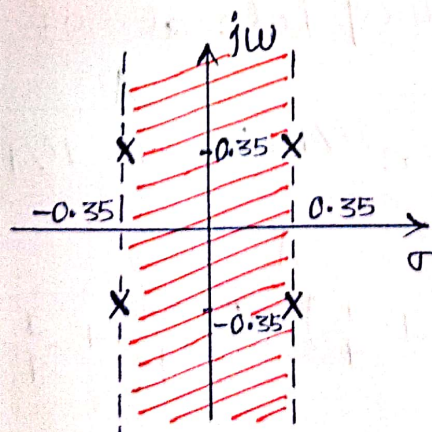
$$\operatorname{Re}\{s\} > 0.35$$

$$-0.35 < \operatorname{Re}\{s\} < 0.35$$

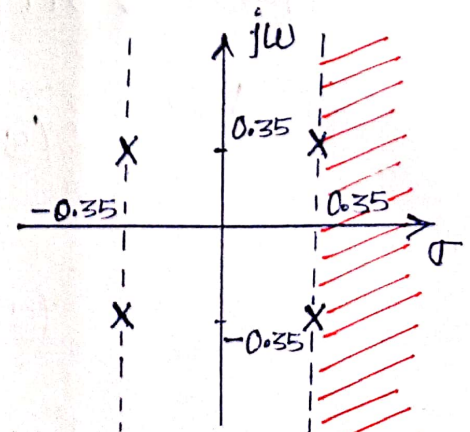
} → (1 MARK)



ROC: $\sigma < -0.35$

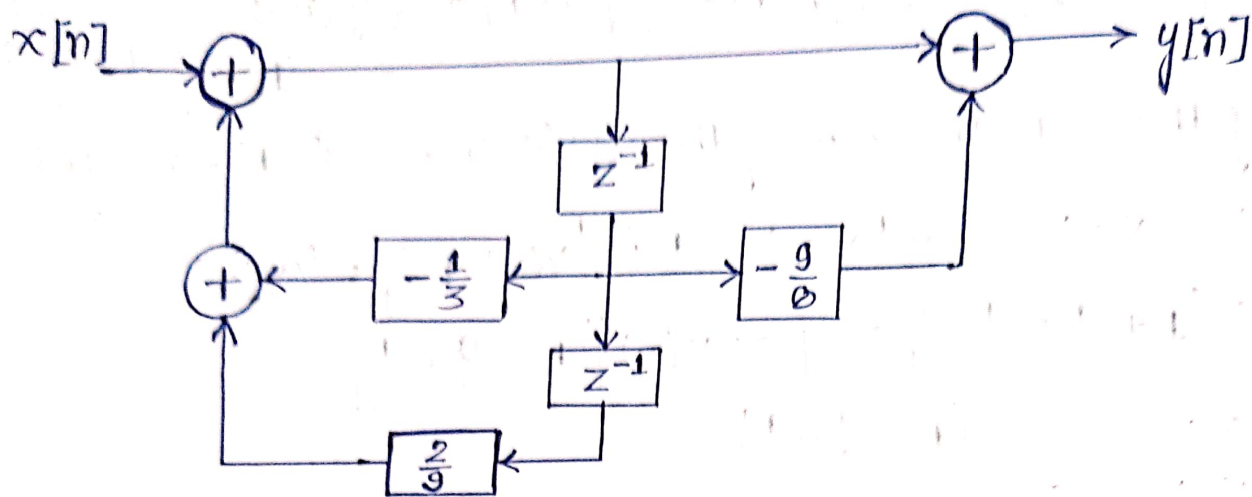


ROC: $-0.35 < \sigma < 0.35$

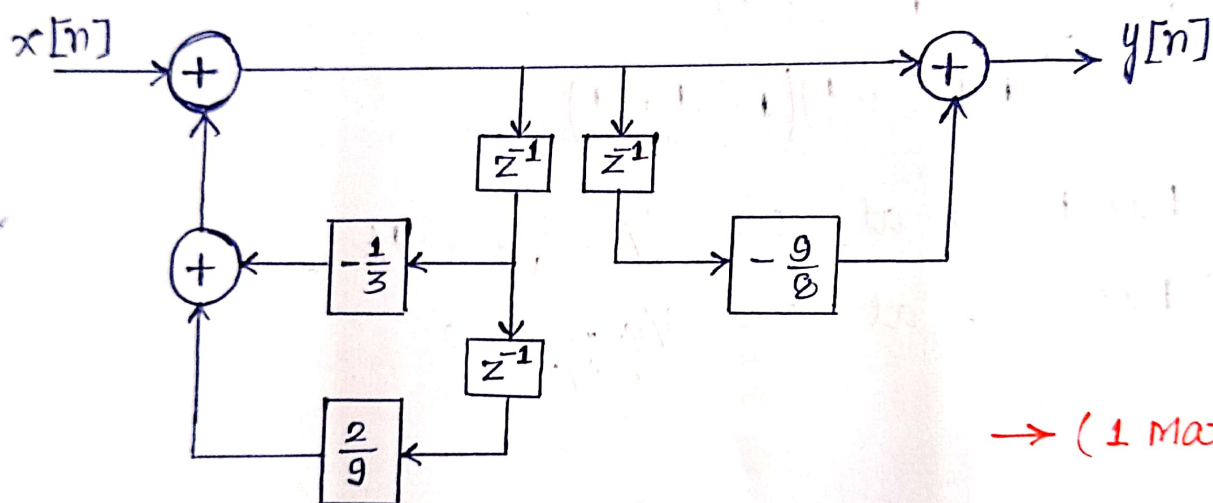


ROC: $\sigma > 0.35$

Sol(6)÷(a) Given block diagram representation,

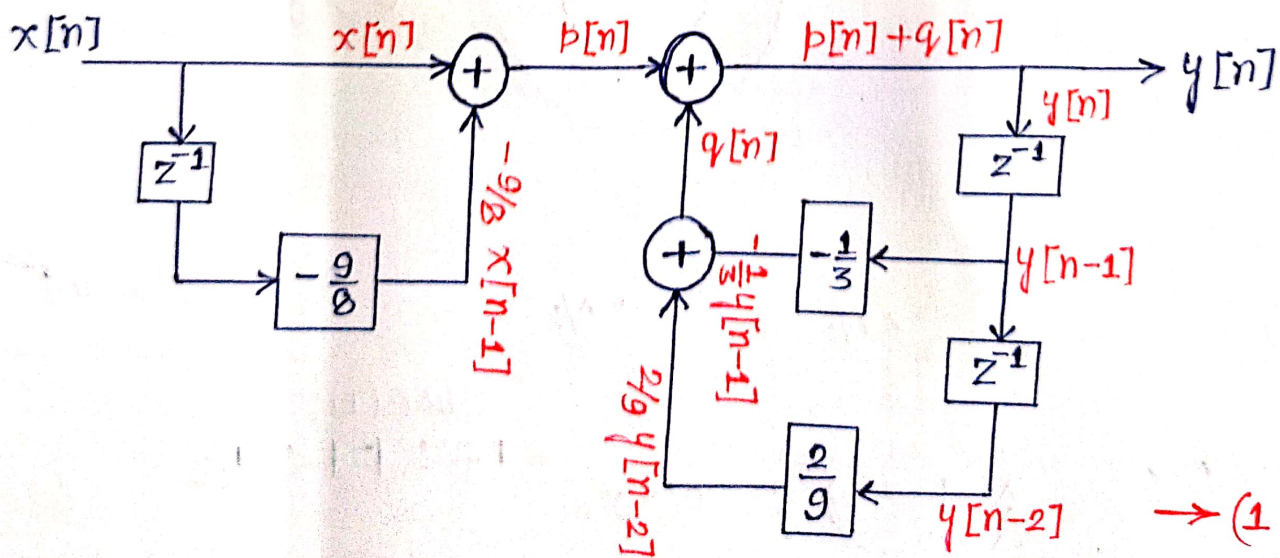


⇓



→ (1 Mark)

⇓



→ (1 Mark)

$$p[n] = x[n] - \frac{9}{8} x[n-1], \quad q[n] = -\frac{1}{3} y[n-1] + \frac{2}{9} y[n-2]$$

$$y[n] = p[n] + q[n]$$

$$y[n] = x[n] - \frac{9}{8}x[n-1] - \frac{1}{3}y[n-1] + \frac{2}{9}y[n-2] \quad \text{--- (1) } \rightarrow (1 \text{ Mark})$$

(b) Taking the z-transform of the difference eqⁿ (1), we get-

$$Y(z) = X(z) - \frac{9}{8}z^{-1}X(z) - \frac{1}{3}z^{-1}Y(z) + \frac{2}{9}z^{-2}Y(z)$$

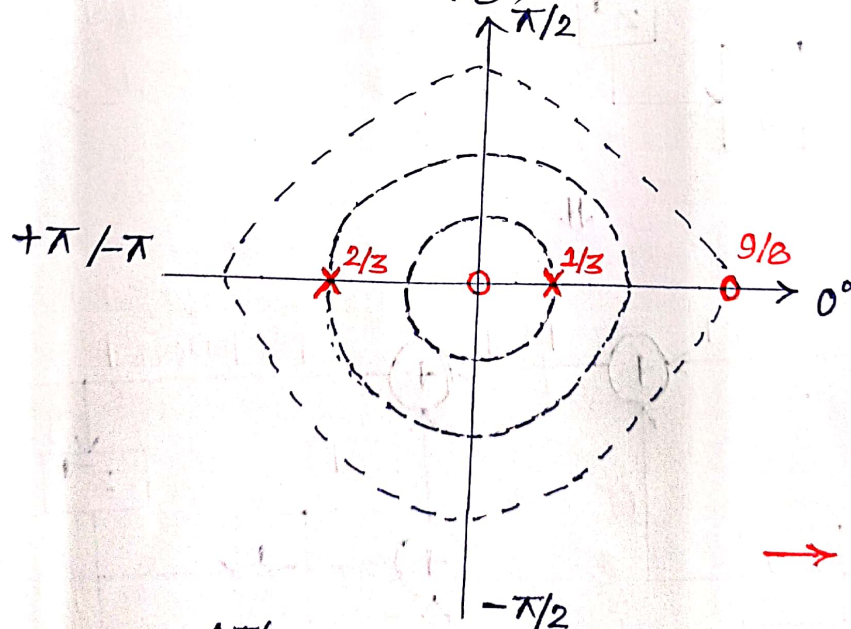
$$Y(z) \left[1 + \frac{1}{3}z^{-1} - \frac{2}{9}z^{-2} \right] = X(z) \left[1 - \frac{9}{8}z^{-1} \right]$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\left(1 - \frac{9}{8}z^{-1} \right)}{\left(1 + \frac{1}{3}z^{-1} - \frac{2}{9}z^{-2} \right)}$$

$$H(z) = \frac{\left(1 - \frac{9}{8}z^{-1} \right)}{\left(1 + \frac{2}{3}z^{-1} \right) \left(1 - \frac{1}{3}z^{-1} \right)} \quad \text{--- (2) } \rightarrow (0.5 \text{ Mark})$$

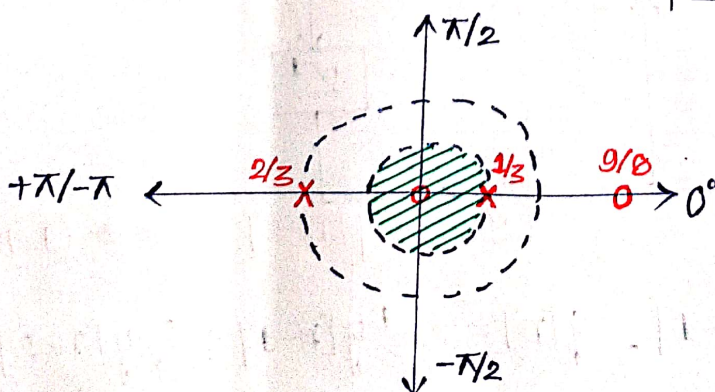
$H(z)$ has poles at $z = -2/3$ & $z = 1/3$

$H(z)$ has zero at $z = 9/8, z = 0$



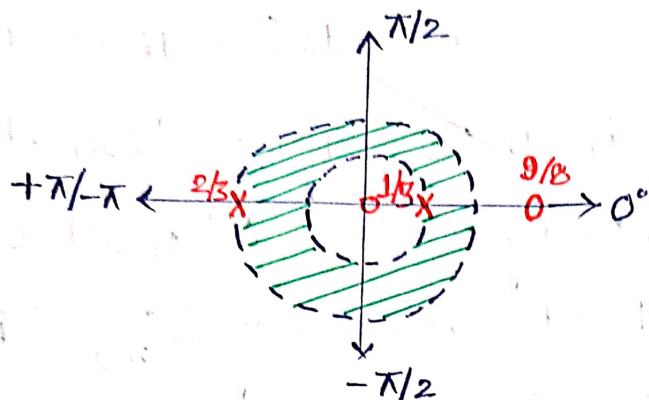
$\rightarrow (0.5 \text{ Mark})$

(c)

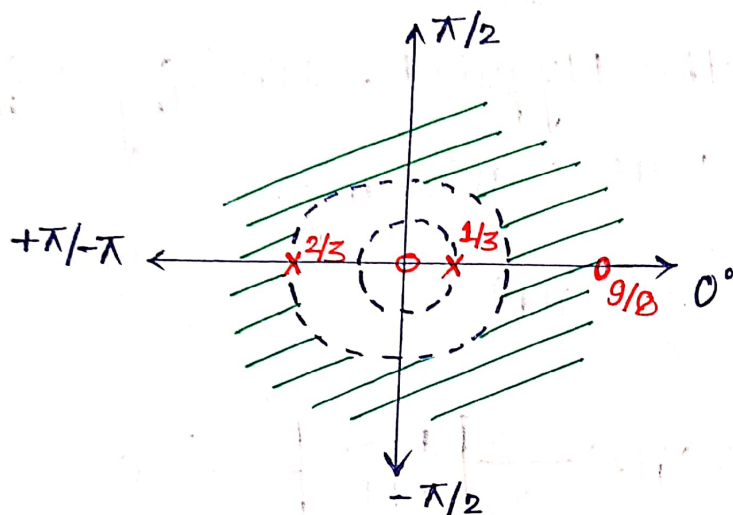


Case (I):
ROC: $|z| < 1/3$

$\rightarrow (0.34 \text{ Mark})$



Case (II):
 ROC: $\frac{1}{3} < |z| < \frac{2}{3}$
 → (0.33 Mark)



Case (III):
 ROC: $|z| > \frac{2}{3}$
 → (0.33 Mark)

(d) For stability of the system, ROC must include unit circle in z-plane. → (0.25 Mark)

For causal system, ROC will be outside circle in z-plane (outer most pole). → (0.25 Mark)

The above two condition satisfied by the case (III).

Hence, required ROC:

By eqⁿ (2), we have-

$$|z| > \frac{2}{3} \rightarrow (0.5 \text{ Mark})$$

$$H(z) = \frac{(1 - \frac{9}{8}z^{-1})}{(1 + \frac{2}{3}z^{-1})(1 - \frac{1}{3}z^{-1})} = \frac{(\frac{43}{24})}{(1 + \frac{2}{3}z^{-1})} - \frac{(\frac{19}{24})}{(1 - \frac{1}{3}z^{-1})}$$

$$\therefore h[n] = (\frac{43}{24})(-\frac{2}{3})^n u[n] - (\frac{19}{24})(\frac{1}{3})^n u[n]$$

→ (0.5 Mark)

Sol(7):

(a) Nyquist Sampling Theorem:

Let $x(t)$ be a band-limited signal with $x(j\omega) = 0$ for $|\omega| > \omega_m$. Then $x(t)$ is uniquely determined by its samples $x(nT)$, $n = 0, \pm 1, \pm 2, \dots$ if

$$\omega_s > 2\omega_m$$

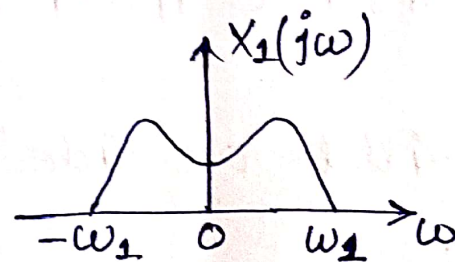
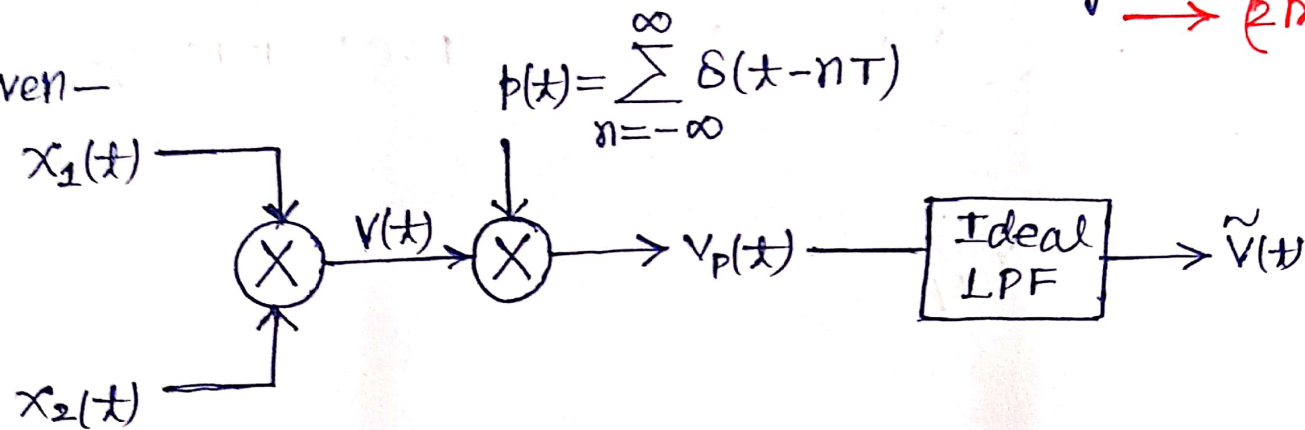
where

$$\omega_s = \frac{2\pi}{T}$$

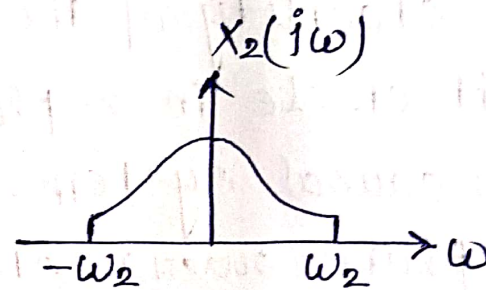
Given these samples, we can reconstruct $x(t)$ by generating a periodic impulse train in which successive impulses have amplitudes that are successive sample values. This impulse train is then processed through an ideal LPF with cutoff freq. greater than ω_m & less than $(\omega_s - \omega_m)$. The resulting output signal will exactly equal $x(t)$.

→ (2M)

(b) Given—



$$X_1(j\omega) = 0, \quad |\omega| \geq \omega_1$$



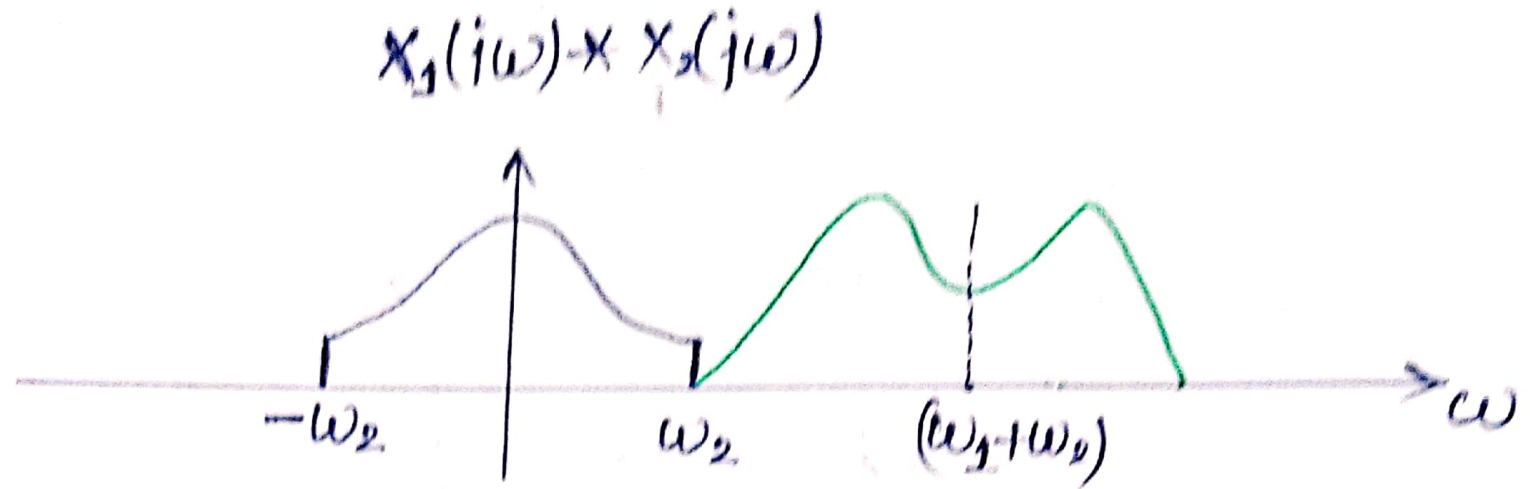
$$X_2(j\omega) = 0, \quad |\omega| \geq \omega_2$$

$$v(t) = x_1(t) \cdot x_2(t) \implies V(j\omega)$$

$$\therefore V(j\omega) = \frac{1}{2\pi} \{ X_1(j\omega) * X_2(j\omega) \} \quad \text{--- (1)}$$

Since $X_1(j\omega) = 0$ for $|\omega| \geq \omega_1$ and $X_2(j\omega) = 0$ for $|\omega| \geq \omega_2$. We may conclude that (from eqⁿ (1)),
 $V(j\omega) = 0$ for $|\omega| \geq (\omega_1 + \omega_2)$

↓ (1 M)



The above instant, after which convolution of $X_1(j\omega)$ & $X_2(j\omega)$ will start zero.

$$\text{Hence, } V(j\omega) = \frac{1}{2\pi} \int X_1(j\omega) \times X_2(j\omega) d\omega$$

$$= 0 \quad \text{for } |\omega| \geq (\omega_1 + \omega_2)$$

→ (1 M)

———— (A)

Consequently, the Nyquist rate for $v(t)$ is -

$$\omega_0 \geq 2(\omega_1 + \omega_2) \quad \text{because, } \omega_{\max} = (\omega_1 + \omega_2) \rightarrow (1M)$$

Therefore, the maximum sampling period which would still allow $v(t)$ to be recovered is -

$$T = \frac{2\pi}{\omega_0} = \frac{2\pi}{2(\omega_1 + \omega_2)} = \frac{\pi}{(\omega_1 + \omega_2)} \rightarrow (2M)$$

(C) Given that - $x_1(t) = \cos(\omega_1 t)$, $f_1 = 2\text{KHz}$
 $x_2(t) = \cos(\omega_2 t)$, $\omega_1 = 4\pi \text{K rad/sec}$
 $f_2 = 5\text{KHz}$
 $\omega_2 = 10\pi \text{K rad/sec}$

As we know that -

$$1 \xrightarrow{FT} 2\pi \delta(\omega)$$

$$e^{j\omega_0 t} \xrightarrow{FT} 2\pi \delta(\omega - \omega_0) \quad \text{--- (2)}$$

$$x_1(t) = \cos(\omega_1 t) = \frac{e^{j\omega_1 t} + e^{-j\omega_1 t}}{2} \xrightarrow{FT} X_1(j\omega)$$

$$X_1(j\omega) = \frac{1}{2} \{ 2\pi \delta(\omega - \omega_1) + 2\pi \delta(\omega + \omega_1) \} \quad (\text{By eqn (2)})$$

$$= \pi \{ \delta(\omega - \omega_1) + \delta(\omega + \omega_1) \} \quad \text{--- (3)}$$

Similarly, $x_2(t) \xrightarrow{FT} X_2(j\omega)$

$$X_2(j\omega) = \pi \{ \delta(\omega - \omega_2) + \delta(\omega + \omega_2) \} \quad \text{--- (4)}$$

$$\therefore V(j\omega) = \frac{1}{2\pi} \{ X_1(j\omega) * X_2(j\omega) \}$$

$$= \frac{1}{2\pi} \left\{ \{ \pi \delta(\omega - \omega_1) + \pi \delta(\omega + \omega_1) \} * \{ \pi \delta(\omega - \omega_2) + \pi \delta(\omega + \omega_2) \} \right\}$$

$$= \frac{1}{2} \left\{ \delta(\omega - \omega_1 - \omega_2) + \delta(\omega + \omega_1 - \omega_2) + \delta(\omega - \omega_1 + \omega_2) + \delta(\omega + \omega_1 + \omega_2) \right\} \quad \text{--- (5)}$$

Given that - $p(t) = \sum_{n=-\infty}^{\infty} \delta(t-nT) = \sum_{k=-\infty}^{\infty} a_k \cdot e^{jk\omega_0 t}$ — (6)

$p(t)$ is periodic with period, T

$\& \omega_0 = 28.4\pi \text{ Krad/sec}$
 $f_0 = 14.2 \text{ KHz}$

$$\therefore a_k = \frac{1}{T} \int_T \delta(t) \cdot e^{-jk\omega_0 t} \cdot dt = \frac{1}{T} = 14.2 \times 10^3$$

$$\therefore p(j\omega) = \sum_{k=-\infty}^{\infty} \frac{1}{T} \cdot 2\pi \delta(\omega - k\omega_0) = 28.4\pi \times 10^3 \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0)$$

— (7)

According to the question,

$$V_p(t) = V(t) \cdot p(t)$$

$$\Rightarrow V_p(j\omega) = \frac{1}{2\pi} \{ V(j\omega) * p(j\omega) \}$$

$$= \frac{1}{2\pi} \left[\frac{1}{2} \{ \delta(\omega - \omega_1 - \omega_2) + \delta(\omega + \omega_1 - \omega_2) + \delta(\omega + \omega_2 - \omega_1) + \delta(\omega + \omega_1 + \omega_2) \} * \left\{ 28.4\pi \times 10^3 \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0) \right\} \right]$$

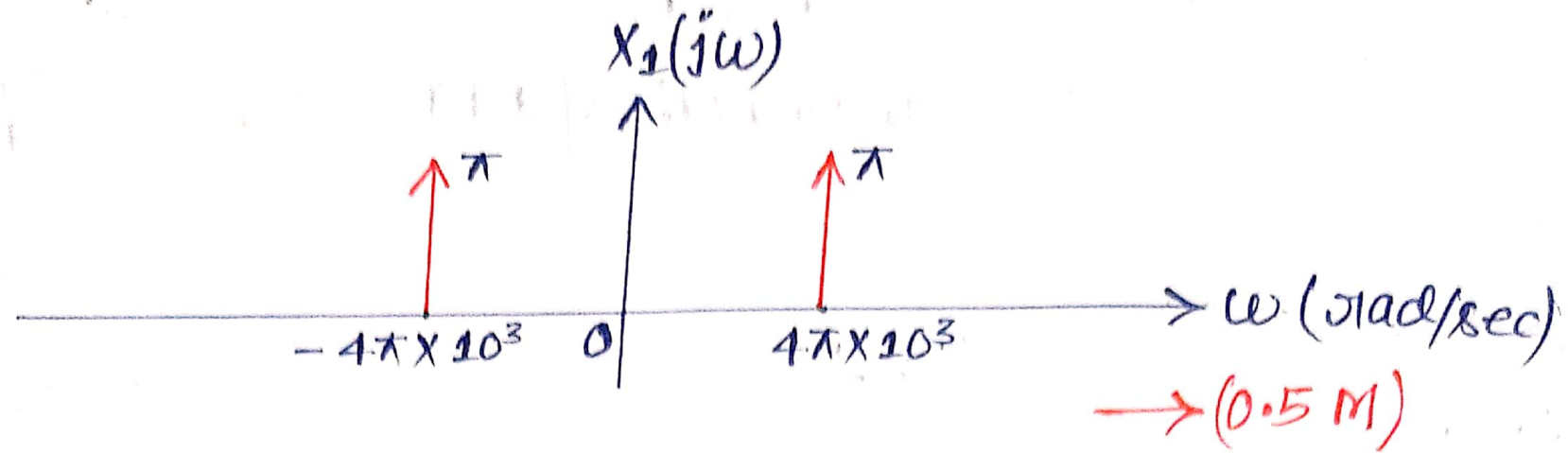
$$= 7.1 \times 10^3 \sum_{k=-\infty}^{\infty} \{ \delta(\omega - k\omega_0 - \omega_1 - \omega_2) + \delta(\omega - k\omega_0 + \omega_1 - \omega_2)$$

$$+ \delta(\omega - k\omega_0 - \omega_1 + \omega_2) + \delta(\omega - k\omega_0 + \omega_1 + \omega_2) \}$$

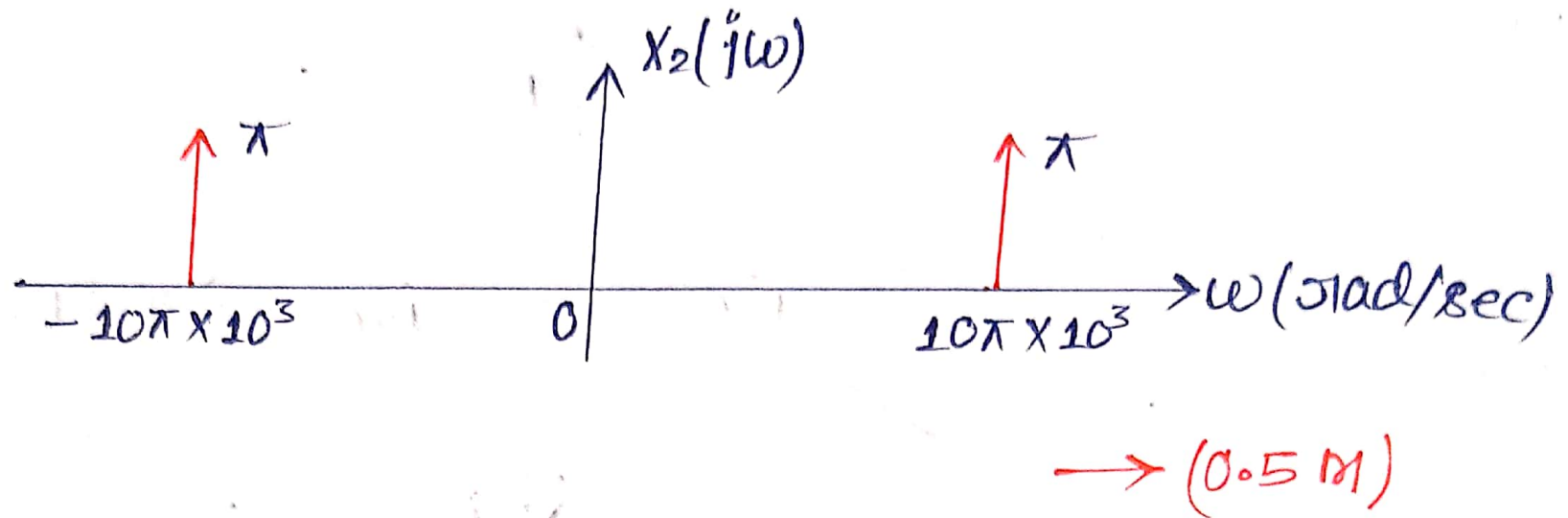
— (8)



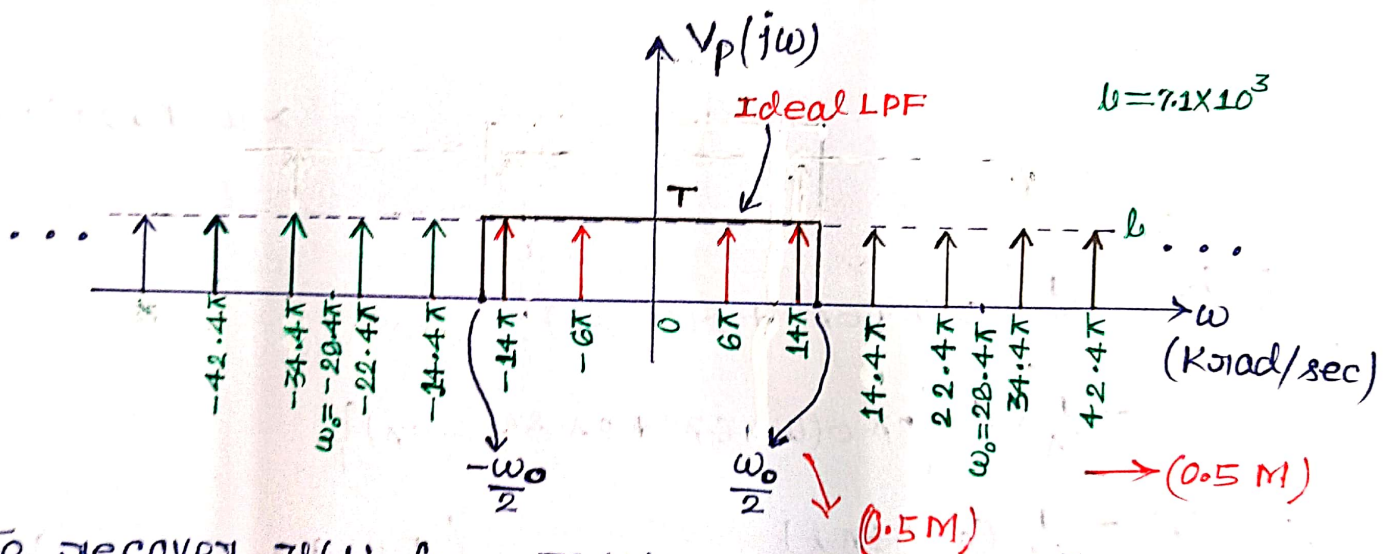
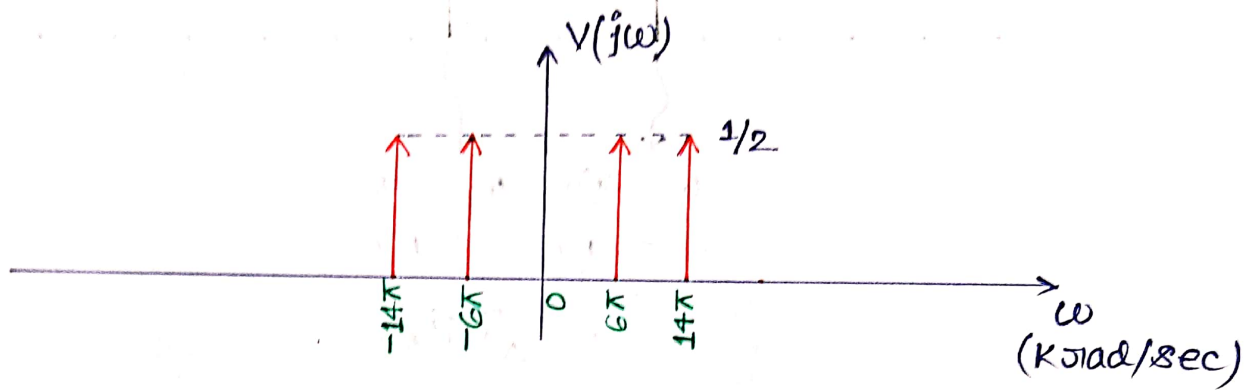
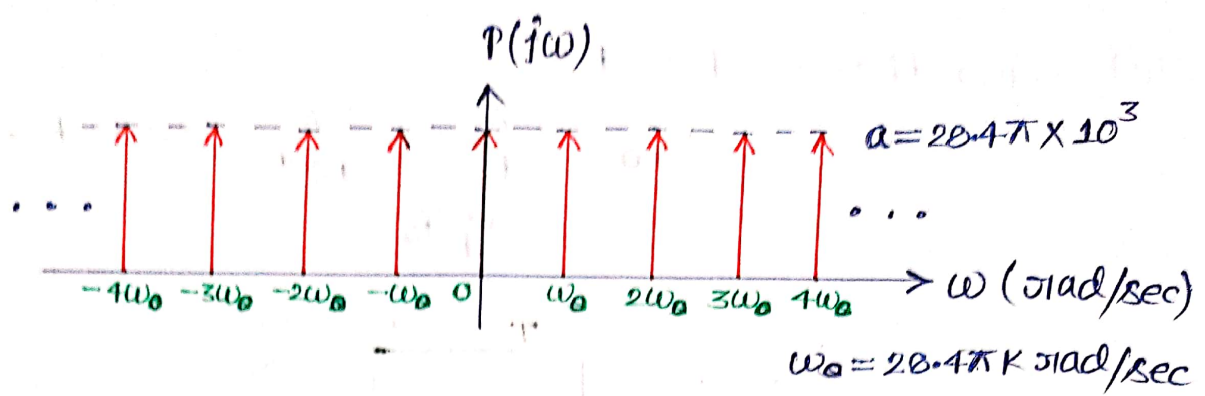
By eqⁿ(3),



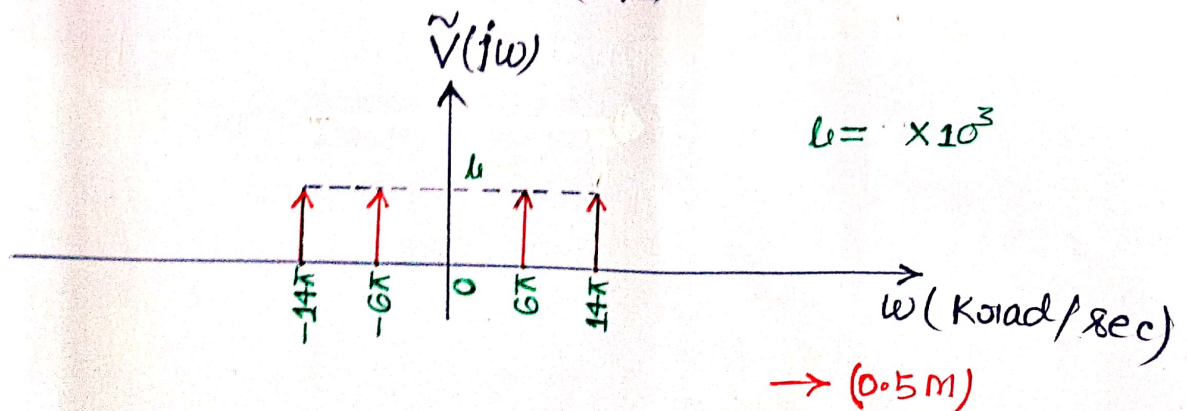
By eqⁿ(4),



By eqⁿ(8),

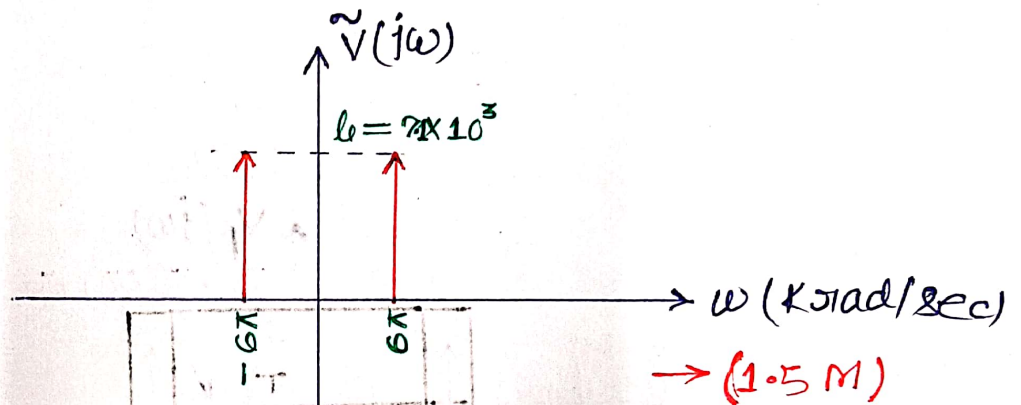
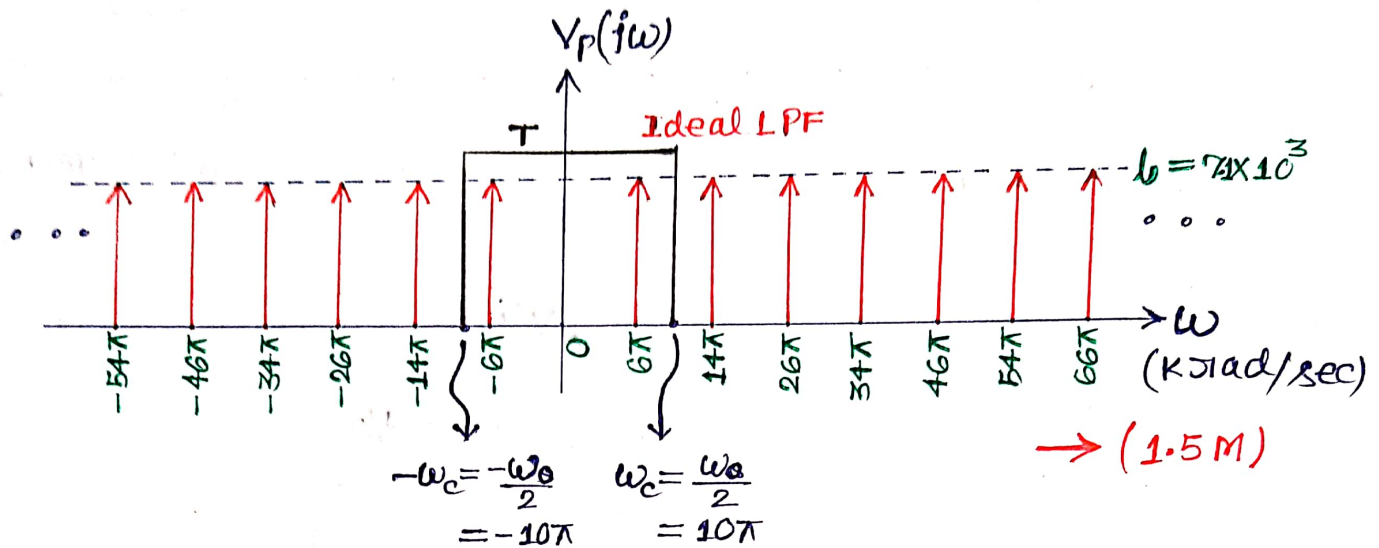


To recover $v(t)$ from $v_p(t)$, $\frac{\omega_0}{2} = 10.2\pi \times 10^3 \text{ rad/sec}$
 $\therefore$ Bandwidth of ideal LPF, $2(\omega_0/2) = 20.4\pi \times 10^3 \text{ rad/sec}$



(d) Given that - $T = 10^{-4}$

$$\therefore \omega_0 = \frac{2\pi}{T} = \frac{2\pi}{10^{-4}} = 20\pi \text{ Krad/sec}$$



$$\tilde{V}(j\omega) = \frac{1}{2} \left\{ \delta(\omega + 6\pi) + \delta(\omega - 6\pi) \right\}$$

$$= \frac{1}{2\pi} \times \frac{1}{2} \left\{ 2\pi \delta(\omega + 6\pi) + 2\pi \delta(\omega - 6\pi) \right\}$$

$$\therefore \tilde{v}(t) = \frac{1}{2\pi} \cos(6\pi t) \quad (\text{By eqn (2) \& eqn (3)})$$

→ (0.5 M)